

# Supplementary material for "Constrained, not identifiable: statistical constraints on the production of the Voynich manuscript text"

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## S1. Exploratory profile: materials and methods

**Parsing.** Page headers of the form `<f1r> <! $Q=A $P=A $F=a $B=1 $I=T $L=A $H=1 ...>` supply quire, illustration section, Currier language and hand. Locus lines `<f1r.1,@P0>` text carry a locus type whose first letter, after stripping the position markers `@ + * = ~ ! / &`, gives the kind: P (paragraph text), L (label), C (circular), R (radial). Loci whose type is prefixed by `&` (labels placed in drawings) are labels. Inline markup: alternatives `[a:o]` take the first reading; extended glyph codes `@nnn`; and `?` mark a token as uncertain, and such tokens are removed; comments `<! ... >` and annotations `{...}` are removed; drawing intrusions `<->` and `<~>` are spaces; paragraph markers `<%>` and `<$>` are removed; fillers `!` are deleted. A `.` is a space (35,750 occurrences) and a `,` an uncertain space (9,316), joined in the baseline. Ligature and capitalised variants are kept as distinct glyphs. Result: 4,130 paragraph-text lines on 207 pages, 32,428 tokens; 1,114 label tokens.

**Token counts.** Three counts of the same paragraph text appear in this paper, and they differ only in preprocessing:

- 32,428 tokens: uncertain spaces joined (the baseline used throughout, and in the pre-specified tests);
- 32,399 tokens: the same after dropping lines with fewer than two tokens, as the profile battery requires;
- 34,886 tokens: uncertain spaces treated as spaces, used only for the basic statistics of Table S1 and as a robustness check (Table S10).

**Line breaks.** For consecutive lines of the same paragraph, the excess MI of the pair (last token of line  $k$ , first token of line  $k+1$ ) is compared with that of adjacent within-line pairs subsampled to the same  $n$  (eight subsamples). For consecutive lines of the same page we also compute the excess MI of all combinations of position classes, and the page specificity  $I(\text{token}; \text{page})$  by position in the line.

**Boundary predictability.** Spaces are removed from each line and a model  $P(\text{boundary} \mid k \text{ glyphs on the left}, k \text{ on the right})$  is fitted on a random half of the lines and evaluated on the other half. We report the information gain  $1 - H(\text{boundary} \mid \text{context})/H(\text{boundary})$  and  $F_1$  at threshold 0.5. The rule-spaced Latin control is stripped of spaces and re-spaced after `m`, `s`, `t` or `y` once the current token has at least three letters.

**Lexicon test.** A within-token glyph model with two glyphs of context and token start and end symbols is fitted on a random half of the tokens, and tokens are sampled from it. We compare the share of held-out real tokens (and types) whose string occurs in the training half with the same share for the sampled tokens. An open inventory defined only by local glyph constraints gives equal shares; a lexicon gives a large gap.

**Entropy profile and residual.** Each line is a string with the space as a symbol. Plug-in conditional entropies  $h(k) = H(x \mid k \text{ preceding symbols})$ ,  $k = 1-5$ , are compared with synthetic twins sampled from Markov models of order 1–3 fitted to the same corpus with the same line lengths; only the comparison with the twin is interpreted, because plug-in  $h(k)$  is biased downward for large  $k$ . The

residual is the lzma rate (preset 9, bits per symbol) of a text minus that of an order-3 twin fitted separately for each combination of Currier language and section. A negative residual means redundancy the drifting automaton does not capture. The automaton's own samples (about  $\pm 0.03$  bit) set the noise floor.

**Generators.** All generators produce text in the line-length, paragraph and page structure of the Voynich paragraph text.

- *Glyph automata.* Counts of order- $k$  contexts over the glyph stream of each line with  $k$  start symbols and an end symbol; sampling per line with the target glyph length of the corresponding real line, the end symbol suppressed until 60% of the target length and sampling stopped at 140%. Section drift: one model per (Currier language, section) group with at least 30 lines, otherwise the global model. Page drift: the next-symbol distribution is  $\lambda$  times the page's own normalised counts plus  $(1 - \lambda)$  times the section's, for contexts seen on the page; this variant sees the page's own text and is in-sample by construction.
- *Table and grille* (after Rugg 2004). Slot decomposition of each attested token by the regular expression  $\text{prefix} = q?(o|y|d|s|ch|sh)?(k|t|p|f|ck|ct|cp|cf|ch|sh|ckh|cth|cph|cfh|l|r|d|s)?$ ,  $\text{midfix} = e\{0,3\}(o|a|y)?$ ,  $\text{suffix} = (i\{0,3\}(n|r|l|m)|d|l|r|s|y|dy|ly|ry|dl|ol|al|or|ar|am|an)?y?$ , with unmatched tokens split into thirds. Per section group a table of 40 columns and  $\max(2, \text{round}(0.9 \times \text{tokens}/40))$  rows, each cell an independent draw of prefix, midfix and suffix from the empirical frequencies. Each page starts at a random cell; a line is the run of cells to the right (wrapping); each line advances one row.
- *Copy and modify* (after Timm and Schinner 2020). Seeds: the 300 most frequent real tokens. Source: line above, same column  $\pm 1$  (0.6); previous token in the line (0.2); random token of the page (0.15); seed (rest). Modification with probability 0.85, a second with 0.25: substitution within the similarity pairs (ch,sh), (k,t), (t,p), (k,f), (e,ee), (ee,eee), (i,ii), (ii,iii), (a,o), (y,dy), (r,l), (n,m), (ol,or), (ar,al), (ain,aiin), (ey,edy) with probability 0.5; otherwise addition or removal of a prefix from  $\{q,o,d,s,y,l,ch,sh\}$  (0.25) or a suffix from  $\{y,dy,n,in,iin,r,l,s,m\}$  (0.25), chosen so that tokens stay around five to six glyphs.
- *Verbose cipher with fixed homophone choice.* Each of 23 Latin letters maps to one to three strings from a pool of 35 Voynich-like units; the homophone is selected by the code of the previous glyph modulo the number of homophones. Spaces after y, n, r, l, m, s once the current token has three glyphs, forced at nine. One independent Latin sentence per line.
- *Ciphertext-feedback substitution.*  $c_t = T[c_{t-1}][p_t]$  with 27 rows, each a random injective map from 23 letters to 27 glyph symbols; the two-glyph variant applies a second table with the intermediate glyph as state. Spaces after y, n, r, l, m at a minimum token length of three and a maximum of nine. Independent sentences per line.
- *Naibbe cipher* (Greshko 2025). Tables from the file `naibbe_tables.csv` of the author's repository (138 unigram, 138 prefix and 138 suffix strings for 23 letters and six tables). Preprocessing: lower case,  $w \rightarrow uu$ ,  $j \rightarrow i$ ,  $k \rightarrow c$ , letters only. The plaintext is cut into one- or two-letter tokens with unigram probability  $17/36$  (the last letter of a line is always a unigram). Deck: 20 alpha, 8 beta1, 8 beta2, 8 beta3, 4 gamma1, 4 gamma2, shuffled and dealt without replacement, reshuffled when exhausted; one card per plaintext letter (variant: per token). Two-letter tokens are  $\text{prefix}(\text{first letter}) + \text{suffix}(\text{second letter})$ . Each ciphertext space is removed with probability 0.03 (variant: none). Plaintext: Latin (Caesar) or Italian (Dante), one independent sentence per line (variant: running text). State-dependent variant: with probability  $w$  the table of the next token is drawn from a fixed random distribution over the six tables conditioned on the last glyph of the previous token (one Dirichlet(c) draw per glyph), otherwise from the deck.

**Drift and hand.** For pairs of pages of the same Currier language we compute the Jensen–Shannon divergence between their glyph-bigram distributions (450-glyph samples, additive smoothing 0.05)

as a function of folio distance, grouped by whether the pages share section and hand. Hands follow Davis (2020) as recorded in the ZL metadata.

**Illustration features and text-image test.** Each page image of folios 1r–58v (free of fold-outs; folio 12 is missing) is rendered at 50 dpi from the Beinecke digital facsimile and cropped by 6% on each side. Pixels are classified relative to the page's median (parchment) colour into green, blue and red–brown. Features per page: the area shares of the three classes, their vertical and horizontal centroids, the mean hue of the green class, the blue and red shares of all coloured pixels, and the total coloured share (standardised). The page-to-folio assignment was verified visually on f26r, f27r and f58r. Text distances are Hellinger distances between relative token-frequency vectors (and, separately, glyph-bigram vectors); image distances are Euclidean. Their Spearman correlation over page pairs is tested by permuting page identities of the image matrix (10,000 permutations), excluding pairs closer than three folios and, for page sets with several hands, permuting within hand. The positive control adds  $k$  pseudo-tokens per page that are deterministic functions of four binary features (blue share, green area, red share, vertical centroid). A secondary analysis predicts the section of a page within one hand from its token or bigram frequencies (leave-one-page-out nearest centroid, excluding folio neighbours within a radius), against the majority rate and the nearest remaining folio; it is confounded by folio position because sections occupy contiguous folio ranges.

**Labels.** For each illustrated section, the share of label tokens whose type occurs on at least one other page of the section is compared with that of equally many randomly drawn paragraph-text tokens of the same pages (or, where a page has too little text, of the same section), averaged over 20 draws.

**Uncertainty.** A page bootstrap with replacement was tried first and rejected: duplicated pages create identical lines and token pairs, which inflates plug-in MI (token MI interval 0.49–0.71 around a point estimate of 0.08) and depresses the hapax share. The leave-one-quire-out jackknife (16 quires) has no duplication and is used throughout.

**Re-segmentation.** Identical byte-pair merges (Sennrich, Haddow and Birch 2016) applied to all corpora leave the controls' ratio  $h_2/h_1$  within 0.72–0.85 at every inventory size, whereas the Voynich ratio rises only from 0.54 to 0.74 after 80 merges (Table S26). The low conditional entropy is not an artefact of a letter being split into several glyphs.

## S2. Exploratory profile: tables

### Basic statistics and the five properties.

Table S1. Basic token- and glyph-level statistics. The Voynich rows treat uncertain spaces as spaces (34,886 tokens); all other analyses join them (32,428 tokens; see S1). Reference corpora are contiguous chunks matched to the Voynich token count (Hebrew: whole text).  $h_1$  = unigram entropy,  $h_2$  = conditional entropy given the previous symbol (bits per symbol, space included). Zipf slope: log-log slope over the 200 most frequent types.

| Corpus                  | Tokens | Types | TTR   | Hapax | Mean length | $h_1$ | $h_2$ | $h_2/h_1$ | Zipf slope |
|-------------------------|--------|-------|-------|-------|-------------|-------|-------|-----------|------------|
| Voynich, paragraph text | 34,886 | 6,875 | 0.197 | 0.67  | 4.96        | 3.87  | 2.10  | 0.54      | −0.78      |
| Currier A               | 11,246 | 3,140 | 0.279 | 0.68  | 4.75        | 3.83  | 2.14  | 0.56      | −0.79      |
| Currier B               | 23,091 | 4,724 | 0.205 | 0.68  | 5.07        | 3.87  | 1.99  | 0.51      | −0.81      |
| Latin (Caesar)          | 34,886 | 8,426 | 0.242 | 0.59  | 6.10        | 4.02  | 3.30  | 0.82      | −0.75      |
| German (Faust)          | 30,959 | 6,221 | 0.201 | 0.63  | 4.78        | 4.13  | 3.19  | 0.77      | −0.95      |
| Italian (Dante)         | 34,886 | 6,574 | 0.188 | 0.62  | 3.92        | 4.01  | 3.15  | 0.79      | −0.94      |
| English (War and Peace) | 34,886 | 4,416 | 0.127 | 0.50  | 4.43        | 4.12  | 3.33  | 0.81      | −0.95      |

|                               |        |       |       |      |      |      |      |      |       |
|-------------------------------|--------|-------|-------|------|------|------|------|------|-------|
| Hebrew (Genesis, consonantal) | 17,789 | 6,319 | 0.355 | 0.65 | 4.41 | 4.13 | 3.53 | 0.85 | -0.65 |
|-------------------------------|--------|-------|-------|------|------|------|------|------|-------|

Table S2. Excess mutual information (bits) between adjacent tokens inside lines, for different projections of the token pair, relative to a within-line permutation null with identical marginals. Line-initial and line-final tokens are excluded. Reference corpora are cut into the Voynich line-length distribution.

| Corpus                          | token → token | last glyph → first glyph | last two → first two | last glyph → token | token → first glyph |
|---------------------------------|---------------|--------------------------|----------------------|--------------------|---------------------|
| Voynich (all)                   | 0.076         | 0.146                    | 0.184                | 0.138              | 0.142               |
| Currier A                       | 0.047         | 0.082                    | 0.109                | 0.076              | 0.077               |
| Currier B                       | 0.081         | 0.187                    | 0.227                | 0.159              | 0.172               |
| all, uncertain spaces as spaces | 0.127         | 0.182                    | 0.235                | 0.174              | 0.191               |
| Latin                           | 0.242         | 0.037                    | 0.212                | 0.234              | 0.167               |
| Italian                         | 0.409         | 0.152                    | 0.422                | 0.337              | 0.356               |
| German                          | 0.216         | 0.056                    | 0.173                | 0.191              | 0.156               |
| Hebrew                          | 0.178         | 0.047                    | 0.332                | 0.212              | 0.218               |

Table S3. Excess mutual information (bits) for the pair (last token of line k, first token of line k+1) within paragraphs, compared with adjacent within-line pairs subsampled to the same n (mean ± s.d. over 8 subsamples). Reference corpora were cut into lines and paragraphs with the Voynich length distributions.

| Corpus                          | n pairs | across line break | adjacent, same n | adjacent inner, same n | across, distance 3 | within, distance 3 |
|---------------------------------|---------|-------------------|------------------|------------------------|--------------------|--------------------|
| Voynich (tokens as transcribed) | 3,820   | 0.005             | 0.062 ± 0.007    | 0.071 ± 0.006          | 0.025              | 0.019 ± 0.005      |
| variant glyphs collapsed        | 3,820   | 0.022             | 0.145 ± 0.015    | 0.162 ± 0.012          | 0.051              | 0.051 ± 0.009      |
| Currier A, collapsed            | 1,408   | 0.019             | 0.052 ± 0.016    | 0.058 ± 0.015          | 0.013              | 0.010 ± 0.013      |
| Currier B, collapsed            | 2,348   | 0.017             | 0.145 ± 0.020    | 0.165 ± 0.010          | 0.016              | 0.039 ± 0.015      |
| Latin                           | 3,488   | 0.077             | 0.070 ± 0.008    | 0.067 ± 0.003          | 0.008              | 0.007 ± 0.004      |
| Italian                         | 3,488   | 0.166             | 0.171 ± 0.012    | 0.166 ± 0.018          | 0.000              | 0.009 ± 0.004      |
| German                          | 3,079   | 0.055             | 0.060 ± 0.010    | 0.062 ± 0.009          | 0.012              | 0.001 ± 0.006      |

Table S4. Predictability of token boundaries from k glyphs on each side (model fitted on half the lines, evaluated on the other half). Gain = 1 - H(boundary | context)/H(boundary). F<sub>1</sub> for predicting a boundary at threshold 0.5.

| Corpus    | boundary rate | gain, k = 1 | F <sub>1</sub> , k = 1 | gain, k = 2 | F <sub>1</sub> , k = 2 | gain, k = 3 |
|-----------|---------------|-------------|------------------------|-------------|------------------------|-------------|
| Voynich   | 0.167         | 73.7%       | 0.85                   | 77.2%       | 0.89                   | 60.2%       |
| Currier A | 0.173         | –           | –                      | 70.4%       | 0.86                   | –           |
| Currier B | 0.166         | –           | –                      | 79.5%       | 0.89                   | –           |
| Latin     | 0.147         | 38.7%       | 0.55                   | 61.6%       | 0.79                   | 35.0%       |
| Italian   | 0.232         | 43.1%       | 0.63                   | 53.6%       | 0.76                   | 19.7%       |
| German    | 0.189         | 44.9%       | 0.66                   | 57.8%       | 0.77                   | 30.4%       |

|                                                       |       |       |      |       |      |       |
|-------------------------------------------------------|-------|-------|------|-------|------|-------|
| Hebrew                                                | 0.206 | 40.1% | 0.57 | 42.3% | 0.65 | 17.8% |
| Latin, spaces removed and re-inserted by a glyph rule | 0.145 | 73.2% | 0.84 | 73.3% | 0.87 | 43.2% |

Table S5. Lexicon test. A within-token glyph model with two glyphs of context is fitted on a random half of the tokens; the share of held-out real tokens (types) whose string occurs in the training half is compared with the same share for tokens sampled from the model.

| Corpus                 | real tokens attested | synthetic tokens attested | real types attested | synthetic types attested | bits/glyph real | bits/glyph synthetic |
|------------------------|----------------------|---------------------------|---------------------|--------------------------|-----------------|----------------------|
| Voynich                | 0.79                 | 0.76                      | 0.36                | 0.30                     | 1.90            | 1.89                 |
| Currier A              | 0.71                 | 0.70                      | 0.34                | 0.32                     | 1.95            | 1.93                 |
| Currier B              | 0.79                 | 0.75                      | 0.36                | 0.31                     | 1.80            | 1.79                 |
| Latin                  | 0.81                 | 0.20                      | 0.48                | 0.03                     | 2.68            | 2.67                 |
| German                 | 0.83                 | 0.34                      | 0.44                | 0.07                     | 2.61            | 2.55                 |
| Italian                | 0.84                 | 0.52                      | 0.44                | 0.09                     | 2.75            | 2.70                 |
| Hebrew                 | 0.68                 | 0.36                      | 0.40                | 0.14                     | 3.00            | 2.81                 |
| Latin, rule-spaced     | 0.62                 | 0.28                      | 0.28                | 0.09                     | 2.74            | 2.76                 |
| Latin, syllable pairs  | 0.65                 | 0.09                      | 0.42                | 0.09                     | 2.99            | 2.94                 |
| Latin, 3-letter chunks | 0.96                 | 0.26                      | 0.82                | 0.25                     | 3.15            | 3.07                 |

Table S6. Conditional entropy  $h(k) = H(x \mid k \text{ previous symbols})$  of the glyph stream (space as a symbol, lines as separate strings), and differences to a synthetic twin sampled from the order-3 model fitted to the same corpus (same number and length of lines). lzma: compressed bits per symbol.

| Corpus             | h(1) | h(2) | h(3) | h(4) | h(5) | h(3) – twin | h(4) – twin | h(5) – twin | lzma | lzma twin |
|--------------------|------|------|------|------|------|-------------|-------------|-------------|------|-----------|
| Voynich            | 2.18 | 1.91 | 1.82 | 1.72 | 1.57 | –0.03       | –0.06       | –0.06       | 2.40 | 2.58      |
| Currier A          | 2.23 | 1.95 | 1.83 | 1.68 | 1.46 | –0.01       | –0.03       | –0.03       | 2.62 | 2.76      |
| Currier B          | 2.06 | 1.78 | 1.70 | 1.60 | 1.46 | –0.02       | –0.04       | –0.05       | 2.33 | 2.46      |
| Latin              | 3.38 | 2.72 | 2.11 | 1.66 | 1.33 | 0.00        | –0.30       | –0.30       | 2.58 | 3.11      |
| Italian            | 3.27 | 2.77 | 2.31 | 1.88 | 1.43 | +0.02       | –0.19       | –0.18       | 3.04 | 3.38      |
| German             | 3.29 | 2.63 | 2.10 | 1.71 | 1.38 | +0.01       | –0.19       | –0.17       | 2.88 | 3.22      |
| Hebrew             | 3.63 | 2.90 | 2.22 | 1.60 | 1.05 | +0.09       | –0.13       | –0.15       | 3.41 | 3.83      |
| Latin, rule-spaced | 3.21 | 2.75 | 2.25 | 1.83 | 1.39 | +0.01       | –0.24       | –0.30       | 2.64 | 3.21      |

### Mechanism battery and robustness.

Table S7. Mechanism battery. All generators produce text in exactly the line and token shape of the Voynich paragraph text. Edge MI and token MI: excess over within-line permutation; line-break MI: (last, first) pair across line breaks; residual: lzma bits per glyph of the text minus that of an order-3 section-drift twin fitted to the same text (negative = redundancy beyond the automaton). The page-drift automaton is fitted in-sample (see text).

| Mechanism      | edge MI | token MI | line-break MI | space gain | hapax | Zipf  | attested real | attested synth. | h(2) | h(4) | residual | I(token; page) |
|----------------|---------|----------|---------------|------------|-------|-------|---------------|-----------------|------|------|----------|----------------|
| Voynich (ZL3b) | 0.146   | 0.078    | 0.005         | 73.7%      | 0.71  | –0.75 | 0.79          | 0.75            | 1.91 | 1.72 | –0.039   | 0.232          |

|                                                          |       |       |        |       |      |       |      |      |      |      |        |        |
|----------------------------------------------------------|-------|-------|--------|-------|------|-------|------|------|------|------|--------|--------|
| Glyph automaton, order 3, section drift                  | 0.167 | 0.050 | 0.005  | 73.8% | 0.71 | -0.77 | 0.79 | 0.76 | 1.92 | 1.75 | +0.033 | 0.134  |
| Glyph automaton, order 4, section drift                  | 0.149 | 0.069 | -0.001 | 74.1% | 0.64 | -0.74 | 0.81 | 0.76 | 1.92 | 1.69 | -0.030 | 0.138  |
| Glyph automaton, order 4, page drift ( $\lambda = 0.5$ ) | 0.158 | 0.076 | 0.004  | 73.8% | 0.63 | -0.74 | 0.81 | 0.75 | 1.92 | 1.70 | -0.052 | 0.215  |
| Latin, running text                                      | 0.034 | 0.245 | 0.076  | 38.9% | 0.59 | -0.76 | 0.80 | 0.20 | 2.72 | 1.65 | -0.397 | 0.062  |
| Latin, rule-spaced                                       | 0.134 | 0.198 | 0.067  | 72.8% | 0.77 | -0.55 | 0.62 | 0.29 | 2.75 | 1.82 | -0.429 | 0.043  |
| Latin, one independent sentence per line                 | 0.035 | 0.646 | 0.159  | 38.7% | 0.35 | -0.76 | 0.86 | 0.22 | 2.70 | 1.57 | -0.993 | 0.050  |
| Table-and-grille generator (after Rugg)                  | 0.006 | 0.613 | 0.041  | 62.8% | 0.41 | -0.51 | 0.73 | 0.55 | 2.58 | 1.98 | -0.766 | 0.049  |
| Copy-and-modify generator (after Timm and Schinner)      | 0.010 | 0.022 | 0.042  | 46.7% | 0.62 | -0.41 | 0.57 | 0.23 | 2.69 | 1.97 | -1.030 | 0.912  |
| Verbose cipher, fixed homophone choice                   | 0.409 | 1.237 | 0.136  | 47.4% | 0.30 | -0.77 | 0.93 | 0.64 | 2.20 | 1.58 | -0.950 | 0.015  |
| Ciphertext-feedback substitution, 1 glyph/letter         | 0.649 | 0.160 | 0.006  | 36.5% | 0.78 | -0.41 | 0.39 | 0.27 | 3.37 | 1.66 | -0.310 | 0.003  |
| Ciphertext-feedback substitution, 2 glyphs/letter        | 0.615 | 0.305 | 0.009  | 38.6% | 0.70 | -0.49 | 0.54 | 0.32 | 2.63 | 1.45 | -0.424 | 0.007  |
| Naibbe cipher, Latin, independent lines                  | 0.003 | 0.055 | 0.002  | 77.4% | 0.49 | -1.00 | 0.85 | 0.72 | 1.90 | 1.70 | -0.065 | 0.007  |
| Naibbe cipher, Latin, running text                       | 0.002 | 0.078 | 0.023  | 77.0% | 0.49 | -1.01 | 0.84 | 0.72 | 1.91 | 1.71 | -0.050 | -0.007 |
| Naibbe cipher, Italian, running text                     | 0.002 | 0.066 | 0.029  | 76.8% | 0.47 | -1.01 | 0.86 | 0.69 | 1.92 | 1.73 | -0.065 | 0.001  |
| Naibbe, state-dependent table choice, $w = 0.6$          | 0.039 | 0.111 | 0.003  | 76.7% | 0.48 | -1.02 | 0.85 | 0.68 | 1.93 | 1.67 | -0.101 | -0.007 |
| Naibbe, state-dependent, $w = 0.9$                       | 0.085 | 0.181 | -0.001 | 76.7% | 0.48 | -1.01 | 0.86 | 0.64 | 1.92 | 1.60 | -0.148 | 0.004  |

|                                           |       |       |       |       |      |       |      |      |      |      |        |       |
|-------------------------------------------|-------|-------|-------|-------|------|-------|------|------|------|------|--------|-------|
| Naibbe, state-<br>dependent, $w$<br>= 1.0 | 0.255 | 0.342 | 0.001 | 75.4% | 0.49 | -0.99 | 0.87 | 0.66 | 1.87 | 1.51 | -0.222 | 0.002 |
|-------------------------------------------|-------|-------|-------|-------|------|-------|------|------|------|------|--------|-------|

Table S8. Page-drift automata: order  $o$ , mixing weight  $\lambda$  of page-level counts against section-level counts.

| Automaton                   | edge<br>MI | token<br>MI | line-<br>break<br>MI | space<br>gain | hapax | Zipf  | attested<br>real | attested<br>synth. | h(2) | h(4) | residual | I(token;<br>page) |
|-----------------------------|------------|-------------|----------------------|---------------|-------|-------|------------------|--------------------|------|------|----------|-------------------|
| order 3, $\lambda$<br>= 0.5 | 0.153      | 0.040       | -0.003               | 73.7%         | 0.70  | -0.78 | 0.80             | 0.75               | 1.92 | 1.75 | +0.011   | 0.187             |
| order 4, $\lambda$<br>= 0.3 | 0.160      | 0.066       | 0.002                | 73.5%         | 0.64  | -0.74 | 0.81             | 0.75               | 1.93 | 1.70 | -0.036   | 0.157             |
| order 4, $\lambda$<br>= 0.5 | 0.158      | 0.076       | 0.004                | 73.8%         | 0.63  | -0.74 | 0.81             | 0.75               | 1.92 | 1.70 | -0.052   | 0.215             |
| order 4, $\lambda$<br>= 0.7 | 0.164      | 0.065       | -0.002               | 74.3%         | 0.64  | -0.74 | 0.81             | 0.75               | 1.92 | 1.69 | -0.084   | 0.279             |
| order 5, $\lambda$<br>= 0.3 | 0.162      | 0.104       | -0.001               | 73.9%         | 0.55  | -0.73 | 0.83             | 0.74               | 1.92 | 1.70 | -0.110   | 0.163             |

Table S9. Naibbe cipher variants: one card per letter or per token; with and without the 3% random space removal; independent sentences per line or running text; Latin or Italian plaintext.

| Variant                                             | edge<br>MI | token<br>MI | line-<br>break<br>MI | space<br>gain | hapax | Zipf  | attested<br>real | attested<br>synth. | h(2) | h(4) | residual | I(token;<br>page) |
|-----------------------------------------------------|------------|-------------|----------------------|---------------|-------|-------|------------------|--------------------|------|------|----------|-------------------|
| Latin,<br>independent<br>lines, card per<br>letter  | 0.003      | 0.055       | 0.002                | 77.4%         | 0.49  | -1.00 | 0.85             | 0.72               | 1.90 | 1.70 | -0.065   | 0.007             |
| Latin,<br>independent<br>lines, card per<br>token   | 0.003      | 0.120       | 0.004                | 77.5%         | 0.44  | -0.75 | 0.95             | 0.53               | 1.88 | 1.60 | -0.198   | -0.001            |
| Latin,<br>independent<br>lines, no space<br>removal | 0.002      | 0.069       | 0.009                | 80.8%         | 0.41  | -1.00 | 0.88             | 0.73               | 1.87 | 1.68 | -0.074   | -0.010            |
| Latin, running<br>text                              | 0.002      | 0.078       | 0.023                | 77.0%         | 0.49  | -1.01 | 0.84             | 0.72               | 1.91 | 1.71 | -0.050   | -0.007            |
| Italian,<br>running text                            | 0.002      | 0.066       | 0.029                | 76.8%         | 0.47  | -1.01 | 0.86             | 0.69               | 1.92 | 1.73 | -0.065   | 0.001             |

Table S10. Robustness to transcription and to the treatment of uncertain spaces.

| Transcription                                                    | tokens | types | edge<br>MI | token<br>MI | line-<br>break<br>MI | space<br>gain | hapax | attested<br>real /<br>synth. | h(2) | h(4) | residual | I(token;<br>page) |
|------------------------------------------------------------------|--------|-------|------------|-------------|----------------------|---------------|-------|------------------------------|------|------|----------|-------------------|
| Zandbergen-<br>Landini,<br>uncertain spaces<br>joined (baseline) | 32,399 | 7,832 | 0.146      | 0.078       | 0.005                | 73.7%         | 0.71  | 0.79 /<br>0.75               | 1.91 | 1.72 | -0.039   | 0.232             |
| Zandbergen-<br>Landini,<br>uncertain spaces<br>as spaces         | 34,859 | 6,863 | 0.182      | 0.128       | 0.004                | 74.0%         | 0.67  | 0.83 /<br>0.81               | 1.89 | 1.71 | -0.044   | 0.243             |
| Takahashi                                                        | 34,384 | 7,054 | 0.165      | 0.116       | 0.011                | 75.2%         | 0.68  | 0.83 /<br>0.79               | 1.89 | 1.71 | -0.039   | 0.245             |

Table S11. Out-of-sample comparison (automaton fitted per section on the odd-numbered pages, evaluated on the even-numbered pages) and the battery restricted to single scribal hands and to the two Currier languages.

| Subset                               | edge MI | token MI | line-break MI | space gain | hapax | Zipf  | attested real / synth. | h(2) | h(4) | I(token; page) |
|--------------------------------------|---------|----------|---------------|------------|-------|-------|------------------------|------|------|----------------|
| Held-out real pages (even)           | 0.150   | 0.034    | -0.001        | 74.1%      | 0.71  | -0.74 | 0.75 / 0.73            | 1.89 | 1.66 | 0.289          |
| Twin sampled from the odd-page model | 0.150   | 0.007    | 0.000         | 73.8%      | 0.71  | -0.77 | 0.74 / 0.73            | 1.92 | 1.70 | 0.168          |
| Hand 1 (112 pages)                   | 0.079   | 0.060    | 0.001         | 70.5%      | 0.71  | -0.77 | 0.71 / 0.71            | 1.93 | 1.65 | 0.129          |
| Hand 2 (46 pages)                    | 0.195   | 0.087    | -0.001        | 77.4%      | 0.70  | -0.84 | 0.78 / 0.74            | 1.72 | 1.50 | 0.237          |
| Hand 3 (31 pages)                    | 0.185   | 0.041    | 0.006         | 74.8%      | 0.72  | -0.77 | 0.71 / 0.69            | 1.79 | 1.56 | 0.182          |
| Currier A                            | 0.082   | 0.049    | 0.007         | 70.6%      | 0.72  | -0.76 | 0.71 / 0.72            | 1.95 | 1.68 | 0.131          |
| Currier B                            | 0.187   | 0.081    | 0.007         | 76.8%      | 0.71  | -0.78 | 0.78 / 0.76            | 1.78 | 1.60 | 0.224          |

Table S12. Leave-one-quire-out jackknife standard errors for the core Voynich statistics (16 quires).

| Statistic                       | estimate | jackknife s.e. | 95% interval     |
|---------------------------------|----------|----------------|------------------|
| edge MI                         | 0.146    | 0.024          | 0.099 to 0.193   |
| token MI                        | 0.081    | 0.024          | 0.034 to 0.127   |
| line-break MI                   | 0.004    | 0.010          | -0.016 to 0.024  |
| adjacent MI at the line-break n | 0.058    | 0.022          | 0.015 to 0.101   |
| space gain (k = 1)              | 0.737    | 0.018          | 0.702 to 0.772   |
| hapax share                     | 0.710    | 0.006          | 0.698 to 0.722   |
| Zipf slope                      | -0.753   | 0.034          | -0.819 to -0.687 |
| attested, real                  | 0.792    | 0.026          | 0.742 to 0.843   |
| attested, synthetic             | 0.754    | 0.025          | 0.704 to 0.803   |
| lexicon gap (real - synthetic)  | 0.039    | 0.021          | -0.003 to 0.080  |
| h(2)                            | 1.905    | 0.047          | 1.812 to 1.998   |
| h(4)                            | 1.717    | 0.025          | 1.669 to 1.765   |

**Drift, hand, illustrations and labels.**

Table S13. Mean Jensen-Shannon divergence between the glyph-bigram distributions of two pages (450-glyph samples) as a function of their folio distance, within the same section and Currier language.

| Pages                                | d = 1-2 | 3-6   | 7-14  | 15-40 | > 40  | other section |
|--------------------------------------|---------|-------|-------|-------|-------|---------------|
| Voynich, Currier A                   | 0.124   | 0.145 | 0.150 | 0.150 | 0.161 | 0.193         |
| Voynich, Currier B                   | 0.126   | 0.137 | 0.134 | 0.147 | 0.147 | 0.169         |
| Section-drift twin, A                | 0.093   | 0.093 | 0.094 | 0.093 | 0.094 | 0.148         |
| Section-drift twin, B                | 0.086   | 0.085 | 0.085 | 0.092 | 0.092 | 0.122         |
| Latin in book order (chunked like A) | 0.183   | 0.188 | 0.195 | 0.192 | 0.192 | 0.193         |

Table S14. Mean Jensen-Shannon divergence between pages grouped by scribal hand (palaeographic assignment of Davis 2020 as recorded in the ZL metadata) and illustration section.

| Pairs                              | same section, same hand | same section, different hand | different section, same hand | different section, different hand |
|------------------------------------|-------------------------|------------------------------|------------------------------|-----------------------------------|
| Currier B, all sections            | 0.135                   | 0.153                        | 0.172                        | 0.168                             |
| Herbal-B, folio distance ≤ 6 pages | 0.138                   | 0.182                        | -                            | -                                 |

|                                        |       |       |   |   |
|----------------------------------------|-------|-------|---|---|
| Herbal-B, folio distance<br>> 20 pages | 0.157 | 0.165 | – | – |
|----------------------------------------|-------|-------|---|---|

Table S15. Mantel tests between text distance and illustration-feature distance across herbal pages (pairs closer than 3 folios excluded).

| Page set                                         | text representation      | Mantel r | p (permutation) | text vs folio distance r (p) |
|--------------------------------------------------|--------------------------|----------|-----------------|------------------------------|
| Herbal-A, hand 1 (85 pages)                      | token frequencies        | –0.055   | 0.79            | +0.115 (0.001)               |
|                                                  | glyph-bigram frequencies | –0.070   | 0.87            | –                            |
| Herbal-B, hands 2 and 5 (25 pages), unstratified | token frequencies        | +0.223   | 0.046           | +0.163 (0.044)               |
| permutation within hand                          | token frequencies        | +0.223   | 0.076           | –                            |
| same-hand pairs only                             | token frequencies        | +0.144   | 0.14            | –                            |
| hand 2 only (20 pages)                           | token frequencies        | +0.143   | 0.14            | –                            |
| hand 2 only                                      | glyph-bigram frequencies | –0.036   | 0.58            | –                            |

Table S16. Sensitivity control: image-determined pseudo-tokens added to each Herbal-A page (mean page length 77 tokens).

| tokens added per page | Mantel r | p     |
|-----------------------|----------|-------|
| 2                     | –0.036   | 0.711 |
| 5                     | +0.053   | 0.209 |
| 10                    | +0.195   | 0.001 |
| 20                    | +0.366   | 0.001 |

Table S17. Herbal-A: Mantel test for each single image feature.

| Image feature                 | Mantel r (tokens) | p    |
|-------------------------------|-------------------|------|
| green area share              | –0.102            | 0.98 |
| blue area share               | –0.049            | 0.75 |
| red-brown area share          | 0.000             | 0.49 |
| green vertical centroid       | –0.085            | 0.94 |
| green horizontal centroid     | –0.006            | 0.54 |
| red vertical centroid         | +0.035            | 0.22 |
| mean green hue                | –0.020            | 0.64 |
| blue share of coloured pixels | –0.059            | 0.80 |
| red share of coloured pixels  | –0.014            | 0.57 |
| total coloured share          | –0.120            | 0.99 |

Table S18. Leave-one-page-out nearest-centroid prediction of the illustration section within a scribal hand, from token frequencies and from glyph-bigram frequencies, excluding pages within the given folio radius; compared with the majority rate and with predicting from the nearest remaining folio.

| Hand (sections)                   | majority rate | excluded radius (folios) | tokens | glyph bigrams | nearest folio only | n   |
|-----------------------------------|---------------|--------------------------|--------|---------------|--------------------|-----|
| Hand 1 (herbal vs pharmaceutical) | 0.86          | 0                        | 0.79   | 0.75          | 0.97               | 111 |

|                               |      |    |      |      |      |     |
|-------------------------------|------|----|------|------|------|-----|
|                               | 0.86 | 5  | 0.79 | 0.75 | 0.84 | 111 |
|                               | 0.86 | 15 | 0.81 | 0.80 | 0.85 | 86  |
| Hand 2 (biological vs herbal) | 0.51 | 0  | 0.74 | 0.95 | 1.00 | 39  |
|                               | 0.51 | 5  | 0.68 | 0.89 | 0.95 | 37  |
|                               | 0.51 | 15 | 0.42 | 0.79 | 0.49 | 19  |
| Hand 3 (stars vs herbal)      | 0.77 | 0  | 0.81 | 0.77 | 0.97 | 31  |
|                               | 0.77 | 5  | 0.77 | 0.77 | 0.71 | 31  |
|                               | 0.77 | 15 | 0.77 | 0.65 | 0.03 | 31  |

Table S19. Recurrence of label tokens across the pages of a section, compared with equally many randomly drawn paragraph-text tokens from the same pages (mean of 20 draws).

| Section        | pages | label tokens | label types | share recurring on another page | null: text tokens of the same pages |
|----------------|-------|--------------|-------------|---------------------------------|-------------------------------------|
| Zodiac         | 12    | 330          | 271         | 0.276                           | 0.363                               |
| Astronomical   | 5     | 142          | 136         | 0.028                           | 0.273                               |
| Pharmaceutical | 15    | 234          | 219         | 0.090                           | 0.364                               |
| Biological     | 11    | 122          | 111         | 0.115                           | 0.342                               |
| Cosmological   | 5     | 196          | 181         | 0.020                           | 0.277                               |

**Neighbour similarity and line structure.** Adjacent inner tokens share their first or last glyphs, or are identical, as often as tokens of the same line in random order (ratios 0.97 to 1.03), and single-glyph edits between neighbours are not directional (largest  $z = 1.8$ ).

Table S20. Share of adjacent inner tokens that share the given feature, observed / within-line permutation null = ratio.

| Corpus    | n pairs | first glyph             | first two               | last glyph              | last two                | last three              | identical               |
|-----------|---------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|-------------------------|
| Voynich   | 19,895  | 0.193 / 0.197<br>= 0.98 | 0.120 / 0.124<br>= 0.97 | 0.311 / 0.314<br>= 0.99 | 0.168 / 0.163<br>= 1.03 | 0.093 / 0.091<br>= 1.02 | 0.011 / 0.011 =<br>0.98 |
| Currier A | 5,661   | 0.210 / 0.212<br>= 0.99 | 0.125 / 0.125<br>= 1.00 | 0.253 / 0.251<br>= 1.01 | 0.132 / 0.130<br>= 1.02 | 0.062 / 0.061<br>= 1.01 | 0.012 / 0.012 =<br>0.94 |
| Currier B | 13,980  | 0.186 / 0.191<br>= 0.98 | 0.118 / 0.124<br>= 0.95 | 0.335 / 0.339<br>= 0.99 | 0.183 / 0.178<br>= 1.03 | 0.106 / 0.103<br>= 1.03 | 0.010 / 0.011 =<br>0.94 |
| Latin     | 22,441  | 0.055 / 0.063<br>= 0.87 | 0.010 / 0.016<br>= 0.62 | 0.150 / 0.135<br>= 1.10 | 0.044 / 0.041<br>= 1.07 | 0.008 / 0.012<br>= 0.68 | 0.000 / 0.002 =<br>0.00 |
| Italian   | 22,441  | 0.046 / 0.061<br>= 0.76 | 0.006 / 0.014<br>= 0.45 | 0.185 / 0.159<br>= 1.16 | 0.013 / 0.017<br>= 0.76 | 0.001 / 0.006<br>= 0.20 | 0.000 / 0.005 =<br>0.09 |
| German    | 19,859  | 0.051 / 0.065<br>= 0.79 | 0.011 / 0.018<br>= 0.60 | 0.120 / 0.124<br>= 0.97 | 0.040 / 0.046<br>= 0.86 | 0.008 / 0.016<br>= 0.48 | 0.002 / 0.008 =<br>0.24 |
| Hebrew    | 11,373  | 0.105 / 0.129<br>= 0.81 | 0.025 / 0.035<br>= 0.73 | 0.114 / 0.119<br>= 0.95 | 0.022 / 0.029<br>= 0.74 | 0.007 / 0.014<br>= 0.48 | 0.001 / 0.006 =<br>0.24 |

Table S21. Single-glyph edits between adjacent inner tokens at edit distance 1 (739 pairs), forward versus inverse counts;  $z = (n_{ab} - n_{ba}) / \sqrt{(n_{ab} + n_{ba})}$ . Share of pairs in edits whose inverse is less than half as frequent: 0.18.

| edit a → b   | count a → b | count b → a | z   |
|--------------|-------------|-------------|-----|
| ins t @mid   | 14          | 6           | 1.8 |
| sub r>l @end | 36          | 23          | 1.7 |
| del d @mid   | 30          | 20          | 1.4 |
| ins i @mid   | 16          | 10          | 1.2 |

|                |    |    |     |
|----------------|----|----|-----|
| del e @mid     | 50 | 40 | 1.1 |
| sub c>s @start | 17 | 13 | 0.7 |
| del o @start   | 18 | 14 | 0.7 |
| sub t>k @mid   | 27 | 24 | 0.4 |
| ins q @start   | 37 | 35 | 0.2 |

Table S22. Similarity of a token to candidate copy sources (Timm and Schinner 2020) and to controls.

| Relation                                   | pairs  | mean edit distance | P(identical) | P(ED ≤ 1) | P(ED ≤ 2) |
|--------------------------------------------|--------|--------------------|--------------|-----------|-----------|
| previous token in the line                 | 30,756 | 4.44               | 0.009        | 0.046     | 0.126     |
| token above (same index)                   | 29,512 | 4.50               | 0.010        | 0.044     | 0.125     |
| closest of three tokens above              | 31,008 | 3.42               | 0.026        | 0.108     | 0.280     |
| token two lines above (same index)         | 27,584 | 4.54               | 0.010        | 0.043     | 0.119     |
| random token of the same page              | 34,859 | 4.61               | 0.012        | 0.038     | 0.107     |
| closest of three random tokens of the page | 33,042 | 3.34               | 0.037        | 0.109     | 0.280     |
| random token of the same Currier language  | 34,859 | 4.75               | 0.005        | 0.025     | 0.085     |

Table S23. Decomposition of the adjacent-token excess MI (global marginal-preserving null) by edit distance between the two tokens.

| Corpus                            | total excess | ED = 0 | ED = 1 | ED = 2 | ED ≥ 3 |
|-----------------------------------|--------------|--------|--------|--------|--------|
| Voynich                           | 0.203        | 0.027  | 0.105  | 0.146  | −0.075 |
| Voynich, variant glyphs collapsed | 0.309        | 0.041  | 0.079  | 0.135  | 0.055  |
| Voynich, strong collapsing        | 0.292        | 0.052  | 0.062  | 0.114  | 0.064  |
| Currier A, collapsed              | 0.150        | 0.040  | 0.080  | 0.150  | −0.120 |
| Currier B, collapsed              | 0.299        | 0.029  | 0.060  | 0.115  | 0.095  |
| Latin                             | 0.262        | −0.003 | 0.000  | 0.022  | 0.243  |
| Latin, words cut to 4 letters     | 0.347        | −0.003 | −0.001 | 0.022  | 0.329  |
| Italian                           | 0.406        | 0.002  | 0.014  | 0.126  | 0.265  |
| German                            | 0.210        | 0.009  | −0.002 | 0.021  | 0.182  |

Table S24. Page specificity of tokens by position in the line (equal n per position).

| Position in line | Voynich I(token; page) | Voynich types/n | Latin I(token; page) | Italian I(token; page) |
|------------------|------------------------|-----------------|----------------------|------------------------|
| first            | 0.098                  | 0.446           | 0.011                | 0.014                  |
| second           | 0.134                  | 0.332           | 0.006                | 0.006                  |
| middle           | 0.113                  | 0.347           | 0.014                | 0.017                  |
| penult           | 0.106                  | 0.361           | 0.018                | 0.003                  |
| last             | 0.055                  | 0.437           | 0.017                | 0.001                  |

Table S25. Excess MI between tokens of consecutive lines of the same page, by position class.

| Pair of consecutive lines           | Voynich | Latin | Italian |
|-------------------------------------|---------|-------|---------|
| last of line k, first of line k + 1 | 0.003   | 0.065 | 0.178   |
| first, first                        | 0.018   | 0.003 | 0.004   |

|                |       |         |         |
|----------------|-------|---------|---------|
| last, last     | 0.022 | − 0.007 | − 0.002 |
| middle, middle | 0.013 | − 0.002 | − 0.006 |
| middle, first  | 0.009 | − 0.007 | 0.006   |
| last, middle   | 0.007 | − 0.002 | 0.002   |

Table S26.  $h_2/h_1$  after a given number of byte-pair merges (inventory size in parentheses).

| merges | Voynich    | German     | Italian    | Latin      | Hebrew     |
|--------|------------|------------|------------|------------|------------|
| 0      | 0.54 (27)  | 0.77 (31)  | 0.79 (35)  | 0.82 (24)  | 0.85 (28)  |
| 5      | 0.58 (32)  | 0.78 (36)  | 0.79 (40)  | 0.82 (29)  | 0.84 (33)  |
| 10     | 0.63 (37)  | 0.77 (41)  | 0.78 (45)  | 0.80 (34)  | 0.83 (38)  |
| 15     | 0.66 (42)  | 0.76 (46)  | 0.78 (49)  | 0.79 (39)  | 0.82 (43)  |
| 20     | 0.68 (47)  | 0.76 (51)  | 0.76 (54)  | 0.78 (44)  | 0.82 (48)  |
| 25     | 0.69 (52)  | 0.75 (56)  | 0.76 (59)  | 0.77 (49)  | 0.81 (53)  |
| 30     | 0.70 (56)  | 0.75 (61)  | 0.76 (64)  | 0.76 (54)  | 0.80 (58)  |
| 40     | 0.72 (66)  | 0.74 (71)  | 0.75 (74)  | 0.75 (64)  | 0.79 (68)  |
| 50     | 0.73 (76)  | 0.74 (81)  | 0.75 (84)  | 0.75 (74)  | 0.79 (78)  |
| 60     | 0.73 (86)  | 0.73 (91)  | 0.75 (94)  | 0.74 (84)  | 0.78 (88)  |
| 80     | 0.74 (105) | 0.74 (110) | 0.74 (114) | 0.72 (104) | 0.77 (108) |

### S3. Deviations and analysis-plan commits

Table S27. Logged deviations from the analysis plans. Full texts in the repository.

| File    | Deviation (all made before the corresponding Voynich statistic was computed)                                                                                                                                                |
|---------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Battery | Gibberish formatting (data handling not foreseen in Section 4).; T6a null computed exactly.; T5 choice of K.; T1 Monte Carlo precision.; T1 null restricted to the page.; T1 estimand made page-conditional.; LATV dropped. |
| E1      | Tuning grid extended.; Position-stratified expectation.; Inference through the steps.                                                                                                                                       |
| E5      | Uncertainty by quire jackknife instead of page bootstrap.                                                                                                                                                                   |
| E2–E7   | none beyond those recorded in PROGRAM.md (E6/E7: none)                                                                                                                                                                      |

Table S28. Commits of the analysis plans and stages. Each plan was committed before the corresponding Voynich outcome was computed. Commit identities were rewritten once before publication to remove an e-mail address; contents and dates are unchanged (mechanism/COMMIT\_MAP.md).

| Stage                      | Commit  | Committed (author date) | Content                |
|----------------------------|---------|-------------------------|------------------------|
| Battery analysis plan      | b67bb47 | 2026-09-23 04:09 +0200  | PREREG.md frozen       |
| Battery Stage A            | f282965 | 2026-09-23 04:25 +0200  | controls only          |
| Battery Stage B and report | 650abbd | 2026-09-23 04:32 +0200  | Voynich                |
| E1 analysis plan           | 889e908 | 2026-09-23 04:41 +0200  |                        |
| E1 calibration             | c507641 | 2026-09-23 04:45 +0200  | before the Voynich run |
| E2 analysis plan           | 0b40d28 | 2026-09-23 04:49 +0200  |                        |
| E3 analysis plan           | ca203c4 | 2026-09-23 04:55 +0200  |                        |
| E4 analysis plan           | a647269 | 2026-09-23 05:07 +0200  |                        |
| E5 analysis plan           | bc6c840 | 2026-09-23 05:10 +0200  |                        |
| E6 analysis plan           | aadd173 | 2026-09-23 05:13 +0200  |                        |
| E7 analysis plan           | 7b002ad | 2026-09-23 05:21 +0200  |                        |
| Red-team checks            | 355263d | 2026-09-23 10:44 +0200  | post hoc               |

## S4. Generator parameters

Table S29. Fitted generator parameters.  $\lambda$  (lam): quire session weight;  $\sigma$  (sig): page tilt;  $\sigma_w$  (sigw): line-to-line drift;  $\rho$ ,  $\tau$ ,  $e$ : recency rate, range, exact share;  $\epsilon$  (eps): slip rate;  $m$ : pruning threshold;  $\theta$  (theta): choice temperature;  $c$ : coinage rate;  $sq$ ,  $sp$ ,  $sw$ : quire, page and line variation of repertoire weights.

| Model              | Fitted parameters (training half)                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------|
| E2 HGR             | lam = 0.25, sig = 0.4, rho = 0.08, tau = 8, e = 0.1, eps = 0                                      |
| E3 HGR2 (fit even) | m = 1, theta = 1.0, eps = 0.08, rho = 0.05, tau = 8, e = 0.1, lam = 0.25, sig = 0.4, sigw = 0.05  |
| E3 HGR2 (fit odd)  | m = 1, theta = 1.0, eps = 0.08, rho = 0.05, tau = 16, e = 0.1, lam = 0.25, sig = 0.4, sigw = 0.05 |
| E6 TR (fit even)   | m = 1, c = 0.1, eps = 0.02, rho = 0.03, tau = 4, e = 0.1, sq = 0.4, sp = 0.4, sw = 0              |

## S5. Held-out statistics for every generator

Table S30. E2: held-out statistics (test: even bifolios).

| Statistic                     | Voynich | full      | no recency | no pruning | no session | plain automaton |
|-------------------------------|---------|-----------|------------|------------|------------|-----------------|
| hapax share †                 | 0.7085  | 0.6022 ✗  | 0.6314 ✗   | 0.6799 ✗   | 0.6275 ✗   | 0.7044 ✓        |
| type-token ratio †            | 0.3012  | 0.2190 ✗  | 0.2110 ✗   | 0.2825 ✓   | 0.2261 ✗   | 0.2914 ✓        |
| Zipf slope                    | −0.7460 | −0.7448 ✓ | −0.7588 ✓  | −0.7537 ✓  | −0.7460 ✓  | −0.7633 ✓       |
| mean word length              | 5.3867  | 5.1717 ✓  | 5.1360 ✓   | 5.2468 ✓   | 5.1550 ✓   | 5.2544 ✓        |
| boundary MI                   | 0.1523  | 0.1328 ✓  | 0.1645 ✓   | 0.1315 ✓   | 0.1325 ✓   | 0.1563 ✓        |
| word-pair MI                  | 0.0549  | 0.0255 ✓  | 0.0582 ✓   | 0.0122 ✓   | 0.0304 ✓   | 0.0340 ✓        |
| distance-2 word MI            | 0.0425  | 0.0233 ✓  | 0.0192 ✓   | 0.0103 ✗   | 0.0095 ✗   | 0.0153 ✗        |
| line-break word MI            | 0.0006  | −0.0014 ✓ | 0.0038 ✓   | 0.0010 ✓   | 0.0040 ✓   | 0.0021 ✓        |
| line-break boundary MI        | 0.0120  | −0.0032 ✓ | 0.0163 ✓   | 0.0052 ✓   | 0.0016 ✓   | −0.0009 ✓       |
| boundary MI, equal n          | 0.1511  | 0.1370 ✓  | 0.1660 ✓   | 0.1246 ✓   | 0.1272 ✓   | 0.1406 ✓        |
| space predictability          | 0.7388  | 0.7847 ✗  | 0.7910 ✗   | 0.7464 ✓   | 0.7772 ✗   | 0.7375 ✓        |
| lexicon test, real words      | 0.7378  | 0.8133 ✗  | 0.8255 ✗   | 0.7487 ✓   | 0.8089 ✗   | 0.7481 ✓        |
| lexicon test, synthetic words | 0.7186  | 0.7657 ✓  | 0.7779 ✓   | 0.7183 ✓   | 0.7613 ✓   | 0.7127 ✓        |
| conditional entropy h2        | 1.8787  | 1.8107 ✓  | 1.7747 ✗   | 1.9186 ✓   | 1.8226 ✓   | 1.9219 ✓        |
| conditional entropy h3        | 1.7779  | 1.7097 ✓  | 1.6695 ✗   | 1.7958 ✓   | 1.7130 ✓   | 1.7954 ✓        |
| conditional entropy h4        | 1.6475  | 1.6415 ✓  | 1.6066 ✓   | 1.6929 ✓   | 1.6385 ✓   | 1.6909 ✓        |
| lzma bits per character       | 2.4440  | 2.3285 ✓  | 2.2877 ✓   | 2.4836 ✓   | 2.3491 ✓   | 2.4943 ✓        |
| page MI †                     | 0.2529  | 0.3048 ✓  | 0.2172 ✓   | 0.1993 ✓   | 0.2168 ✓   | 0.1731 ✗        |
| T1 line-break ratio           | 0.0809  | −0.1428 ✓ | −0.0347 ✓  | 0.2131 ✓   | −0.0763 ✓  | −0.0125 ✓       |
| T2a final form, margin        | 0.0942  | 0.1346 ✓  | 0.1441 ✓   | 0.1230 ✓   | 0.1268 ✓   | 0.1357 ✓        |
| T2a final form, paragraph end | 0.0373  | 0.1653 ✗  | 0.1770 ✗   | 0.1653 ✗   | 0.1691 ✗   | 0.1614 ✗        |
| T2b line-fill slack           | 1.9622  | 9.4848 ✗  | 9.3461 ✗   | 10.5469 ✗  | 9.6503 ✗   | 10.3487 ✗       |
| T2c dittography               | 0.6098  | 1.9992 ✓  | 1.3019 ✓   | 1.6935 ✓   | 1.4505 ✓   | 0.9910 ✓        |
| T3a zero replication †        | 0.0384  | 0.0403 ✓  | 0.0391 ✓   | 0.0518 ✗   | 0.0473 ✗   | 0.0551 ✗        |
| T3b slot consistency          | 0.8683  | 0.8777 ✓  | 0.8835 ✓   | 0.8604 ✓   | 0.8768 ✓   | 0.8576 ✓        |
| T3b monotone share            | 0.5471  | 0.5828 ✓  | 0.5943 ✓   | 0.5541 ✓   | 0.5804 ✓   | 0.5532 ✓        |
| T3c final-position entropy    | 0.6299  | 0.6535 ✓  | 0.6534 ✓   | 0.6593 ✓   | 0.6540 ✓   | 0.6665 ✓        |
| T3c initial-position entropy  | 0.8734  | 0.8734 ✓  | 0.8672 ✓   | 0.8786 ✓   | 0.8725 ✓   | 0.8806 ✓        |
| T4 periodicity                | 0.0000  | 0.0000 ✓  | 0.0000 ✓   | 0.0000 ✓   | 0.0000 ✓   | 0.0000 ✓        |

|                              |         |           |           |           |           |           |
|------------------------------|---------|-----------|-----------|-----------|-----------|-----------|
| T5 drift shape               | 0.7808  | 0.8672 ✓  | 0.6696 ✓  | 0.6003 ✓  | 0.5872 ✓  | 0.5068 ✓  |
| T6a recency, near repeats †  | 0.1399  | 0.1451 ✓  | −0.0081 ✗ | 0.1058 ✓  | 0.0923 ✗  | −0.0130 ✗ |
| T6a recency, exact repeats † | 0.2324  | 0.1925 ✓  | 0.0050 ✗  | 0.2737 ✓  | 0.3201 ✓  | −0.0224 ✓ |
| T6b within-page drift †      | −0.0082 | −0.0016 ✗ | −0.0010 ✗ | −0.0015 ✗ | −0.0017 ✗ | −0.0006 ✗ |
| T7b deviant rate †           | 26.8523 | 31.6494 ✓ | 35.6074 ✓ | 47.0364 ✓ | 37.9130 ✓ | 44.4175 ✓ |
| T7b deviant recurrence †     | 0.0047  | 0.1868 ✗  | 0.1458 ✗  | 0.1214 ✗  | 0.1520 ✗  | 0.0906 ✗  |
| T7b slip share               | 0.2870  | 0.3955 ✓  | 0.4417 ✗  | 0.2873 ✓  | 0.4148 ✓  | 0.2741 ✓  |
| E1 line step                 | 0.0591  | 0.0024 ✓  | 0.0419 ✓  | 0.0091 ✓  | −0.0004 ✓ | 0.0070 ✓  |
| E1 paragraph step            | −0.0549 | −0.0440 ✓ | −0.0018 ✓ | −0.0164 ✓ | −0.0035 ✓ | −0.0103 ✓ |
| E1 within-line decay         | 0.0608  | 0.0002 ✓  | 0.0061 ✓  | 0.0036 ✓  | −0.0168 ✓ | −0.0154 ✓ |
| E1 within-paragraph decay    | 0.0447  | 0.0445 ✓  | −0.0080 ✓ | 0.0290 ✓  | 0.0036 ✓  | −0.0448 ✓ |

† fitting target on the training half. ✓ within tolerance  $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$ .

Table S31. E3 primary direction (fit even, test odd).

| Statistic                     | Voynich | full      | no margin rule | no drift  | no slips etc. | E2 class  |
|-------------------------------|---------|-----------|----------------|-----------|---------------|-----------|
| hapax share †                 | 0.7199  | 0.7227 ✓  | 0.6800 ✗       | 0.7173 ✓  | 0.6246 ✗      | 0.5970 ✗  |
| type-token ratio †            | 0.2871  | 0.3223 ✓  | 0.2323 ✓       | 0.3145 ✓  | 0.2065 ✗      | 0.1871 ✗  |
| Zipf slope                    | −0.7499 | −0.7776 ✓ | −0.7831 ✓      | −0.7783 ✓ | −0.7772 ✓     | −0.7929 ✓ |
| mean word length              | 5.2924  | 5.4708 ✓  | 5.2914 ✓       | 5.4364 ✓  | 5.3135 ✓      | 5.2861 ✓  |
| boundary MI                   | 0.1388  | 0.1264 ✓  | 0.1428 ✓       | 0.1369 ✓  | 0.1498 ✓      | 0.1343 ✓  |
| word-pair MI                  | 0.0594  | −0.0388 ✗ | 0.0345 ✓       | −0.0615 ✗ | −0.0024 ✓     | 0.0452 ✓  |
| distance-2 word MI            | 0.0464  | 0.0168 ✓  | 0.0260 ✓       | 0.0161 ✓  | 0.0392 ✓      | 0.0413 ✓  |
| line-break word MI            | 0.0002  | 0.0047 ✓  | 0.0031 ✓       | 0.0027 ✓  | 0.0173 ✓      | 0.0042 ✓  |
| line-break boundary MI        | 0.0078  | 0.0059 ✓  | 0.0114 ✓       | 0.0057 ✓  | 0.0061 ✓      | 0.0099 ✓  |
| boundary MI, equal n          | 0.1480  | 0.1405 ✓  | 0.1341 ✓       | 0.1483 ✓  | 0.1407 ✓      | 0.1366 ✓  |
| space predictability          | 0.7343  | 0.6962 ✓  | 0.7540 ✓       | 0.7009 ✓  | 0.7816 ✗      | 0.7896 ✗  |
| lexicon test, real words      | 0.7455  | 0.7103 ✓  | 0.7992 ✓       | 0.7175 ✓  | 0.8277 ✓      | 0.8444 ✗  |
| lexicon test, synthetic words | 0.7251  | 0.6753 ✗  | 0.7492 ✓       | 0.6759 ✗  | 0.7617 ✓      | 0.7762 ✓  |
| conditional entropy h2        | 1.9100  | 1.9981 ✓  | 1.8345 ✓       | 1.9888 ✓  | 1.7647 ✗      | 1.7471 ✗  |
| conditional entropy h3        | 1.8075  | 1.8477 ✓  | 1.7161 ✓       | 1.8397 ✓  | 1.6595 ✗      | 1.6535 ✗  |
| conditional entropy h4        | 1.6769  | 1.7241 ✓  | 1.6352 ✓       | 1.7190 ✓  | 1.6023 ✗      | 1.5997 ✗  |
| lzma bits per character       | 2.4581  | 2.5279 ✓  | 2.3318 ✓       | 2.5210 ✓  | 2.2590 ✓      | 2.2254 ✗  |
| page MI †                     | 0.3195  | 0.2646 ✓  | 0.2450 ✓       | 0.2530 ✓  | 0.3493 ✓      | 0.3620 ✓  |
| T1 line-break ratio           | 0.3137  | −0.1355 ✓ | 0.0043 ✓       | −0.1009 ✓ | 0.0489 ✓      | −0.0051 ✓ |
| T2a final form, margin        | 0.0789  | 0.0464 ✓  | 0.0212 ✗       | 0.0480 ✓  | 0.0566 ✓      | 0.0180 ✗  |
| T2a final form, paragraph end | 0.0428  | 0.0538 ✓  | 0.0167 ✗       | 0.0578 ✓  | 0.0587 ✓      | 0.0206 ✓  |
| T2b line-fill slack           | 2.2694  | 2.6287 ✗  | 8.0667 ✗       | 2.6204 ✗  | 2.7059 ✗      | 8.1219 ✗  |
| T2c dittography               | 0.9501  | 1.3470 ✓  | 1.6969 ✓       | 2.1210 ✗  | 1.3516 ✓      | 1.7792 ✓  |
| T3a zero replication †        | 0.0415  | 0.0574 ✗  | 0.0547 ✗       | 0.0621 ✗  | 0.0313 ✗      | 0.0298 ✗  |
| T3b slot consistency          | 0.8684  | 0.8540 ✓  | 0.8745 ✓       | 0.8542 ✓  | 0.8828 ✓      | 0.8848 ✓  |
| T3b monotone share            | 0.5607  | 0.5277 ✓  | 0.5785 ✓       | 0.5353 ✓  | 0.5936 ✓      | 0.6047 ✓  |
| T3c final-position entropy    | 0.6217  | 0.6402 ✓  | 0.6147 ✓       | 0.6358 ✓  | 0.5886 ✓      | 0.5839 ✓  |
| T3c initial-position entropy  | 0.8812  | 0.8772 ✓  | 0.8692 ✓       | 0.8809 ✓  | 0.8638 ✓      | 0.8631 ✓  |
| T4 periodicity                | 0.0000  | 0.0000 ✓  | 0.0000 ✓       | 0.0000 ✓  | 0.0000 ✓      | 0.0000 ✓  |
| T5 drift shape                | 0.8021  | 0.8980 ✓  | 0.6265 ✓       | 0.8263 ✓  | 0.7591 ✓      | 0.7922 ✓  |

|                              |         |           |           |           |           |           |
|------------------------------|---------|-----------|-----------|-----------|-----------|-----------|
| T6a recency, near repeats †  | 0.1635  | 0.1168 ✓  | 0.1155 ✓  | 0.1290 ✓  | 0.0478 ✗  | 0.1141 ✓  |
| T6a recency, exact repeats † | 0.2042  | 0.2052 ✓  | 0.0855 ✓  | 0.1749 ✓  | 0.0136 ✗  | 0.1911 ✓  |
| T6b within-page drift †      | -0.0084 | -0.0019 ✗ | -0.0020 ✗ | -0.0012 ✗ | -0.0016 ✗ | -0.0021 ✗ |
| T7b deviant rate †           | 23.4977 | 49.0845 ✗ | 31.9529 ✓ | 50.6217 ✗ | 22.9400 ✓ | 18.7890 ✓ |
| T7b deviant recurrence †     | 0.0086  | 0.0681 ✗  | 0.0487 ✗  | 0.0642 ✗  | 0.1837 ✗  | 0.1836 ✗  |
| T7b slip share               | 0.2723  | 0.2960 ✓  | 0.4209 ✗  | 0.2797 ✓  | 0.4257 ✗  | 0.4163 ✗  |
| E1 line step                 | 0.0655  | 0.0034 ✓  | 0.0199 ✓  | 0.0189 ✓  | 0.0096 ✓  | 0.0123 ✓  |
| E1 paragraph step            | -0.0244 | -0.0318 ✓ | 0.0055 ✓  | -0.0034 ✓ | -0.0054 ✓ | -0.0012 ✓ |
| E1 within-line decay         | 0.0692  | 0.0165 ✓  | 0.0066 ✓  | 0.0020 ✓  | 0.0075 ✓  | -0.0005 ✓ |
| E1 within-paragraph decay    | 0.0817  | 0.0353 ✓  | 0.0435 ✓  | 0.0234 ✓  | 0.0009 ✓  | 0.0298 ✓  |

† fitting target on the training half. ✓ within tolerance  $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$ .

Table S32. E6 primary direction (fit even, test odd).

| Statistic                        | Voynich | two-route    | no<br>coinage | no repertoire<br>variation | no<br>recency | no boundary<br>rule |
|----------------------------------|---------|--------------|---------------|----------------------------|---------------|---------------------|
| hapax share †                    | 0.7199  | 0.6070 ✗     | 0.3152 ✗      | 0.6781 ✗                   | 0.6526 ✗      | 0.6047 ✗            |
| type-token ratio †               | 0.2871  | 0.1336 ✗     | 0.0640 ✗      | 0.2890 ✓                   | 0.1877 ✗      | 0.1327 ✗            |
| Zipf slope                       | -0.7499 | -0.7634<br>✓ | -0.7907<br>✓  | -0.7242 ✓                  | -0.7807<br>✓  | -0.7716 ✓           |
| mean word length                 | 5.2924  | 5.0055 ✗     | 4.8110 ✗      | 5.5328 ✗                   | 5.1737 ✓      | 4.9951 ✗            |
| boundary MI                      | 0.1388  | 0.1256 ✓     | 0.1238 ✓      | 0.1287 ✓                   | 0.1393 ✓      | 0.0005 ✗            |
| word-pair MI                     | 0.0594  | -0.0032<br>✓ | 0.0313 ✓      | -0.0468 ✗                  | -0.0151 ✗     | -0.0262 ✗           |
| distance-2 word MI               | 0.0464  | 0.0503 ✓     | 0.0746 ✓      | 0.0125 ✓                   | 0.0450 ✓      | 0.0458 ✓            |
| line-break word MI               | 0.0002  | 0.0322 ✓     | 0.0536 ✗      | 0.0182 ✓                   | 0.0272 ✓      | 0.0407 ✗            |
| line-break boundary MI           | 0.0078  | 0.0139 ✓     | 0.0189 ✓      | 0.0180 ✓                   | 0.0189 ✓      | 0.0195 ✓            |
| boundary MI, equal n             | 0.1480  | 0.1365 ✓     | 0.1387 ✓      | 0.1434 ✓                   | 0.1449 ✓      | 0.0041 ✓            |
| space predictability             | 0.7343  | 0.8110 ✗     | 0.8484 ✗      | 0.7342 ✓                   | 0.7830 ✗      | 0.8276 ✗            |
| lexicon test, real words         | 0.7455  | 0.8978 ✗     | 0.9677 ✗      | 0.7491 ✓                   | 0.8471 ✗      | 0.9015 ✗            |
| lexicon test, synthetic words    | 0.7251  | 0.8069 ✗     | 0.8473 ✗      | 0.6896 ✓                   | 0.7681 ✓      | 0.8068 ✗            |
| conditional entropy h2           | 1.9100  | 1.7184 ✗     | 1.6104 ✗      | 1.9271 ✓                   | 1.7833 ✓      | 1.7405 ✗            |
| conditional entropy h3           | 1.8075  | 1.6332 ✗     | 1.5380 ✗      | 1.8163 ✓                   | 1.6911 ✗      | 1.6564 ✗            |
| conditional entropy h4           | 1.6769  | 1.5513 ✗     | 1.4717 ✗      | 1.7150 ✓                   | 1.6022 ✓      | 1.5721 ✗            |
| lzma bits per character          | 2.4581  | 2.2290 ✗     | 2.0911 ✗      | 2.4963 ✓                   | 2.3219 ✓      | 2.2539 ✓            |
| page MI †                        | 0.3195  | 0.4378 ✓     | 0.4760 ✗      | 0.1916 ✗                   | 0.4043 ✓      | 0.3943 ✓            |
| T1 line-break ratio              | 0.3137  | -0.0369<br>✓ | 0.2971 ✓      | 0.0502 ✓                   | -0.0398<br>✓  | undef. ✗            |
| T2a final form, margin           | 0.0789  | 0.1025 ✓     | 0.1033 ✓      | 0.0932 ✓                   | 0.1019 ✓      | 0.0996 ✓            |
| T2a final form, paragraph<br>end | 0.0428  | 0.0461 ✓     | 0.0512 ✓      | 0.0678 ✓                   | 0.0594 ✓      | 0.0565 ✓            |
| T2b line-fill slack              | 2.2694  | 2.7781 ✗     | 2.6571 ✗      | 2.7373 ✗                   | 2.5448 ✗      | 2.7543 ✗            |
| T2c dittography                  | 0.9501  | 1.1848 ✓     | 1.2145 ✓      | 1.1697 ✓                   | 0.9520 ✓      | 1.2275 ✓            |
| T3a zero replication †           | 0.0415  | 0.0353 ✓     | 0.0285 ✗      | 0.0464 ✓                   | 0.0377 ✓      | 0.0330 ✓            |
| T3b slot consistency             | 0.8684  | 0.9086 ✗     | 0.9284 ✗      | 0.8616 ✓                   | 0.8913 ✓      | 0.9071 ✗            |
| T3b monotone share               | 0.5607  | 0.6645 ✗     | 0.7103 ✗      | 0.5279 ✓                   | 0.6164 ✓      | 0.6506 ✗            |
| T3c final-position entropy       | 0.6217  | 0.6148 ✓     | 0.6017 ✓      | 0.6468 ✓                   | 0.6263 ✓      | 0.6149 ✓            |
| T3c initial-position entropy     | 0.8812  | 0.8595 ✓     | 0.8541 ✓      | 0.8748 ✓                   | 0.8645 ✓      | 0.8689 ✓            |

|                              |         |           |           |           |           |           |
|------------------------------|---------|-----------|-----------|-----------|-----------|-----------|
| T4 periodicity               | 0.0000  | 0.0000 ✓  | 0.0000 ✓  | 0.0000 ✓  | 0.0000 ✓  | 0.0000 ✓  |
| T5 drift shape               | 0.8021  | 0.4761 ✓  | 0.3850 ✗  | 0.5684 ✓  | 0.5275 ✓  | 0.4666 ✓  |
| T6a recency, near repeats †  | 0.1635  | 0.0641 ✗  | 0.0368 ✗  | 0.0134 ✗  | −0.0262 ✗ | 0.0488 ✗  |
| T6a recency, exact repeats † | 0.2042  | −0.0005 ✗ | −0.0052 ✗ | 0.1216 ✓  | −0.0339 ✗ | 0.0655 ✗  |
| T6b within-page drift †      | −0.0084 | −0.0002 ✗ | −0.0010 ✗ | −0.0008 ✗ | −0.0006 ✗ | −0.0008 ✗ |
| T7b deviant rate †           | 23.4977 | 27.3205 ✓ | 20.4694 ✓ | 38.1832 ✓ | 28.4477 ✓ | 26.9757 ✓ |
| T7b deviant recurrence †     | 0.0086  | 0.0467 ✗  | 0.1503 ✗  | 0.0942 ✗  | 0.0602 ✗  | 0.0499 ✗  |
| T7b slip share               | 0.2723  | 0.4817 ✗  | 0.8110 ✗  | 0.2929 ✓  | 0.3348 ✓  | 0.4602 ✗  |
| E1 line step                 | 0.0655  | 0.0330 ✓  | 0.0103 ✓  | 0.0055 ✓  | 0.0130 ✓  | 0.0381 ✓  |
| E1 paragraph step            | −0.0244 | 0.0341 ✓  | −0.0033 ✓ | 0.0250 ✓  | −0.0117 ✓ | 0.0118 ✓  |
| E1 within-line decay         | 0.0692  | 0.0018 ✓  | 0.0074 ✓  | −0.0079 ✓ | 0.0063 ✓  | 0.0070 ✓  |
| E1 within-paragraph decay    | 0.0817  | 0.0209 ✓  | 0.0033 ✓  | 0.0348 ✓  | −0.0171 ✗ | 0.0092 ✓  |

† fitting target on the training half. ✓ within tolerance  $2\sqrt{(\text{SE}^2 + \text{SD}^2)} + 1\%$ .

## S6. Drift and burstiness

Table S33. E5: page clustering index B by frequency band and gradient G with quire-jackknife 95% intervals.

| Corpus              | G                    | B 2–4 | B 5–19 | B 20–99 | B ≥ 100 |
|---------------------|----------------------|-------|--------|---------|---------|
| Voynich             | +0.24 (−0.10, 0.59)  | 2.02  | 1.83   | 1.58    | 1.36    |
| Voynich (Takahashi) | +0.19 (−0.28, 0.66)  | 1.91  | 1.87   | 1.58    | 1.34    |
| Latin               | +0.74 (0.57, 0.92)   | 2.95  | 2.29   | 1.40    | 1.07    |
| Italian             | +0.78 (0.47, 1.09)   | 2.59  | 1.56   | 1.19    | 1.02    |
| German              | +1.06 (0.81, 1.30)   | 4.89  | 2.96   | 1.70    | 1.13    |
| English             | +1.08 (0.93, 1.22)   | 5.11  | 2.61   | 1.74    | 1.22    |
| VB-run              | +0.66 (0.49, 0.82)   | 2.23  | 1.48   | 1.15    | 1.03    |
| NAIB-run            | +0.07 (−0.33, 0.46)  | 1.09  | 1.04   | 1.02    | 1.00    |
| LDRIIFT             | −0.19 (−0.52, 0.14)  | 2.22  | 3.02   | 2.67    | 2.12    |
| REP                 | −0.60 (−1.14, −0.05) | 1.53  | 2.06   | 2.77    | 2.40    |
| HGR2                | +0.97 (0.49, 1.44)   | 3.68  | 1.70   | 1.40    | 1.31    |
| MK-sec              | +0.03 (−0.28, 0.34)  | 1.10  | 1.15   | 1.07    | 1.03    |

Table S34. E4: drift slopes of line similarity over line distance 2–8 with page-bootstrap 95% intervals.

| Corpus   | L (lexical)                | S (sublexical)             |
|----------|----------------------------|----------------------------|
| Voynich  | −0.0142 (−0.0275, −0.0022) | −0.0052 (−0.0085, −0.0023) |
| Latin    | −0.0115 (−0.0257, 0.0040)  | −0.0028 (−0.0045, −0.0011) |
| Italian  | −0.0147 (−0.0255, −0.0026) | −0.0121 (−0.0153, −0.0088) |
| German   | −0.0252 (−0.0402, −0.0116) | −0.0043 (−0.0070, −0.0015) |
| English  | −0.0128 (−0.0206, −0.0042) | +0.0003 (−0.0031, 0.0040)  |
| VB-run   | −0.0092 (−0.0190, −0.0001) | +0.0002 (−0.0020, 0.0022)  |
| NAIB-run | +0.0076 (−0.0075, 0.0242)  | −0.0015 (−0.0043, 0.0012)  |
| GDRIFT   | −0.0517 (−0.0660, −0.0385) | −0.0148 (−0.0176, −0.0112) |
| LDRIIFT  | −0.0398 (−0.0498, −0.0291) | +0.0007 (−0.0023, 0.0033)  |
| MK-sec   | −0.0042 (−0.0151, 0.0064)  | −0.0007 (−0.0030, 0.0016)  |

## S7. Revision analyses (R1–R4)

These analyses answer a review of an earlier version. Their criteria were written in mechanism/REVISION\_PLAN.md and committed before computation; R1 is labelled post hoc.

Table S35. R1 (post hoc): each gibberish writer against 50 contiguous Voynich stretches of the same length within one hand. Percentile: share of Voynich stretches with slot consistency at or below the writer's. Verdict by the rule fixed in REVISION\_PLAN.md: supported.

| Writer | Tokens | T3b writer | Voynich T3b, median (5th pct.) | Percentile of writer | T3a writer | Voynich T3a median |
|--------|--------|------------|--------------------------------|----------------------|------------|--------------------|
| DA_01  | 215    | 0.713      | 0.872 (0.811)                  | 0%                   | 0.093      | 0.048              |
| DA_02  | 356    | 0.669      | 0.857 (0.828)                  | 0%                   | 0.052      | 0.035              |
| DC_01  | 364    | 0.739      | 0.864 (0.828)                  | 0%                   | 0.088      | 0.034              |
| DC_03  | 251    | 0.717      | 0.857 (0.817)                  | 0%                   | 0.071      | 0.042              |
| DC_04  | 270    | 0.675      | 0.870 (0.825)                  | 0%                   | 0.129      | 0.038              |
| DC_05  | 200    | 0.626      | 0.851 (0.820)                  | 0%                   | 0.138      | 0.047              |
| DC_06  | 372    | 0.652      | 0.876 (0.832)                  | 0%                   | 0.068      | 0.035              |
| DC_08  | 247    | 0.691      | 0.853 (0.812)                  | 0%                   | 0.046      | 0.040              |
| DC_09  | 218    | 0.666      | 0.854 (0.822)                  | 0%                   | 0.163      | 0.042              |
| DC_10  | 221    | 0.694      | 0.858 (0.818)                  | 0%                   | 0.068      | 0.039              |
| DC_11  | 223    | 0.696      | 0.864 (0.822)                  | 0%                   | 0.120      | 0.038              |
| DC_12  | 207    | 0.648      | 0.856 (0.826)                  | 0%                   | 0.131      | 0.039              |
| DC_15  | 480    | 0.656      | 0.867 (0.838)                  | 0%                   | 0.099      | 0.033              |
| DC_16  | 338    | 0.621      | 0.863 (0.831)                  | 0%                   | 0.384      | 0.038              |
| DC_18  | 421    | 0.652      | 0.857 (0.826)                  | 0%                   | 0.046      | 0.033              |
| DC_20  | 274    | 0.687      | 0.884 (0.829)                  | 0%                   | 0.054      | 0.036              |
| DC_21  | 322    | 0.765      | 0.861 (0.830)                  | 0%                   | 0.063      | 0.040              |
| DC_22  | 150    | 0.618      | 0.855 (0.811)                  | 0%                   | 0.537      | 0.043              |
| DC_25  | 157    | 0.685      | 0.849 (0.809)                  | 0%                   | 0.259      | 0.047              |
| DC_26  | 228    | 0.605      | 0.865 (0.818)                  | 0%                   | 1.250      | 0.043              |
| DC_27  | 332    | 0.643      | 0.875 (0.838)                  | 0%                   | 0.187      | 0.035              |
| DC_28  | 215    | 0.678      | 0.861 (0.824)                  | 0%                   | 0.037      | 0.043              |
| DC_29  | 184    | 0.666      | 0.851 (0.806)                  | 0%                   | 0.328      | 0.038              |
| DC_30  | 322    | 0.699      | 0.872 (0.829)                  | 0%                   | 0.156      | 0.041              |
| DC_31  | 355    | 0.638      | 0.869 (0.838)                  | 0%                   | 0.241      | 0.036              |
| DC_32  | 281    | 0.644      | 0.864 (0.829)                  | 0%                   | 0.094      | 0.035              |
| DC_33  | 188    | 0.778      | 0.857 (0.814)                  | 0%                   | 0.048      | 0.042              |
| DC_34  | 276    | 0.750      | 0.863 (0.824)                  | 0%                   | 0.394      | 0.039              |
| DC_35  | 267    | 0.630      | 0.870 (0.825)                  | 0%                   | 0.454      | 0.038              |
| DC_36  | 199    | 0.567      | 0.863 (0.820)                  | 0%                   | 1.359      | 0.041              |
| DC_37  | 317    | 0.606      | 0.876 (0.833)                  | 0%                   | 0.088      | 0.036              |
| DC_38  | 292    | 0.707      | 0.878 (0.828)                  | 0%                   | 0.045      | 0.037              |
| DC_39  | 312    | 0.659      | 0.854 (0.831)                  | 0%                   | 0.154      | 0.039              |
| DC_40  | 330    | 0.642      | 0.874 (0.825)                  | 0%                   | 0.137      | 0.039              |
| DC_41  | 150    | 0.671      | 0.851 (0.813)                  | 0%                   | 0.113      | 0.046              |
| DC_42  | 342    | 0.716      | 0.874 (0.832)                  | 0%                   | 0.076      | 0.034              |

Table S36. R2: our Naibbe reimplementation enciphering the author's tokenised plaintext (Pliny, *Natural History* 16), against the author's published ciphertext ([github.com/greshko/naibbe-cipher](https://github.com/greshko/naibbe-cipher), commit f2675ec). Also: one-letter token share 0.486 (expected 0.472); 100.0% of 34,764 ciphertext tokens are valid table outputs for their plaintext token; table shares within 0.014 of the deck proportions. Validated by the pre-specified rule: yes.

| Statistic                    | Author's ciphertext | Our reimplementation (5 seeds) | Within tolerance |
|------------------------------|---------------------|--------------------------------|------------------|
| hapax share                  | 0.5046              | 0.5184 $\pm$ 0.0031            | no               |
| type count                   | 6,286               | 6,282 $\pm$ 30                 | yes              |
| mean word length             | 5.3885              | 5.3905 $\pm$ 0.0054            | yes              |
| conditional entropy h2       | 1.9139              | 1.8966 $\pm$ 0.0025            | yes              |
| boundary MI                  | 0.0010              | 0.0018 $\pm$ 0.0010            | yes              |
| word-pair MI                 | 0.0653              | 0.0614 $\pm$ 0.0089            | yes              |
| T3c final-position entropy   | 0.6530              | 0.6486 $\pm$ 0.0018            | yes              |
| T3c initial-position entropy | 0.8748              | 0.8655 $\pm$ 0.0007            | yes              |

Table S37. R3: the statistics behind the Naibbe exclusion, computed on the author's own ciphertext cut as running text into the Voynich page and line shape (32,399 tokens). Criterion (REVISION\_PLAN.md): for T6a, boundary MI and page MI the author's output lies outside the Voynich interval on the same side as our reimplementation. Met: yes.

| Statistic                       | Author's ciphertext       | Our reimplementation | Voynich (95% interval)     |
|---------------------------------|---------------------------|----------------------|----------------------------|
| T6a near recency                | 0.000                     | -0.021               | 0.153 (0.112, 0.198)       |
| boundary MI                     | 0.0010                    | 0.0019               | 0.146 (0.099, 0.193)       |
| page MI                         | 0.000                     | -0.007               | 0.239 (0.120, 0.358)       |
| hapax share                     | 0.50                      | 0.49                 | 0.71                       |
| E4 sublexical drift S           | -0.0019 (-0.0042, 0.0006) | -0.0015              | -0.0052 (-0.0085, -0.0023) |
| T7b recurrence of deviant forms | 1.3%                      | 3.8%                 | 1.1%                       |

Table S38. R4: means over 20 randomly driven and 20 message-driven (lzma-compressed plaintext) runs for each statistic, with Welch p-values (Fisher for T4) and Holm-adjusted values.

| Statistic                     | E7 random | E7 message | p (Holm)     | E8 random | E8 message | p (Holm)     |
|-------------------------------|-----------|------------|--------------|-----------|------------|--------------|
| hapax share                   | 0.7118    | 0.7121     | 0.839 (1.00) | 0.7347    | 0.7355     | 0.468 (1.00) |
| type-token ratio              | 0.2380    | 0.2378     | 0.650 (1.00) | 0.2941    | 0.2958     | 0.075 (1.00) |
| Zipf slope                    | -0.7725   | -0.7733    | 0.706 (1.00) | -0.7600   | -0.7611    | 0.620 (1.00) |
| mean word length              | 5.2765    | 5.2737     | 0.412 (1.00) | 5.4652    | 5.4665     | 0.861 (1.00) |
| boundary MI                   | 0.1585    | 0.1574     | 0.467 (1.00) | 0.1299    | 0.1303     | 0.733 (1.00) |
| word-pair MI                  | 0.0454    | 0.0504     | 0.129 (1.00) | -0.0413   | -0.0472    | 0.115 (1.00) |
| distance-2 word MI            | 0.0255    | 0.0226     | 0.093 (1.00) | 0.0173    | 0.0170     | 0.804 (1.00) |
| line-break word MI            | 0.0012    | 0.0013     | 0.951 (1.00) | 0.0031    | 0.0028     | 0.752 (1.00) |
| line-break boundary MI        | 0.0012    | 0.0022     | 0.448 (1.00) | 0.0010    | 0.0024     | 0.186 (1.00) |
| boundary MI, equal n          | 0.1615    | 0.1619     | 0.907 (1.00) | 0.1373    | 0.1314     | 0.072 (1.00) |
| space predictability          | 0.7416    | 0.7417     | 0.895 (1.00) | 0.6941    | 0.6930     | 0.452 (1.00) |
| lexicon test, real words      | 0.7897    | 0.7906     | 0.294 (1.00) | 0.7345    | 0.7351     | 0.578 (1.00) |
| lexicon test, synthetic words | 0.7587    | 0.7584     | 0.743 (1.00) | 0.7010    | 0.6977     | 0.051 (1.00) |
| conditional entropy h2        | 1.9203    | 1.9197     | 0.622 (1.00) | 2.0434    | 2.0470     | 0.138 (1.00) |
| conditional entropy h3        | 1.8234    | 1.8235     | 0.980 (1.00) | 1.9207    | 1.9244     | 0.085 (1.00) |
| conditional entropy h4        | 1.7497    | 1.7501     | 0.574 (1.00) | 1.8138    | 1.8181     | 0.039 (1.00) |
| lzma bits per character       | 2.4370    | 2.4361     | 0.422 (1.00) | 2.5412    | 2.5459     | 0.085 (1.00) |
| page MI                       | 0.1240    | 0.1226     | 0.478 (1.00) | 0.1799    | 0.1772     | 0.385 (1.00) |
| T1 line-break ratio           | -0.0288   | -0.0077    | 0.473 (1.00) | -0.0723   | 0.0330     | 0.007 (0.29) |
| T2a final form, margin        | 0.1352    | 0.1392     | 0.044 (1.00) | 0.0576    | 0.0580     | 0.788 (1.00) |
| T2a final form, paragraph end | 0.1728    | 0.1742     | 0.723 (1.00) | 0.0550    | 0.0565     | 0.524 (1.00) |

|                              |         |         |              |         |         |              |
|------------------------------|---------|---------|--------------|---------|---------|--------------|
| T2b line-fill slack          | 9.0476  | 9.2420  | 0.047 (1.00) | 2.5410  | 2.5614  | 0.466 (1.00) |
| T2c dittography              | 0.9744  | 0.9849  | 0.935 (1.00) | 1.5533  | 1.7134  | 0.194 (1.00) |
| T3a zero replication         | 0.0532  | 0.0535  | 0.472 (1.00) | 0.0638  | 0.0634  | 0.415 (1.00) |
| T3b slot consistency         | 0.8598  | 0.8606  | 0.020 (0.78) | 0.8467  | 0.8459  | 0.184 (1.00) |
| T3b monotone share           | 0.5594  | 0.5605  | 0.289 (1.00) | 0.5213  | 0.5188  | 0.139 (1.00) |
| T3c final-position entropy   | 0.6676  | 0.6670  | 0.352 (1.00) | 0.6455  | 0.6487  | 0.029 (1.00) |
| T3c initial-position entropy | 0.8783  | 0.8772  | 0.046 (1.00) | 0.8866  | 0.8851  | 0.155 (1.00) |
| T4 periodicity               | 0.0000  | 0.0000  | 1.000 (1.00) | 0.0000  | 0.0000  | 1.000 (1.00) |
| T5 drift shape               | 0.5909  | 0.5927  | 0.888 (1.00) | 0.8368  | 0.8499  | 0.343 (1.00) |
| T6a recency, near repeats    | −0.0082 | −0.0076 | 0.917 (1.00) | 0.0878  | 0.1018  | 0.042 (1.00) |
| T6a recency, exact repeats   | −0.0080 | 0.0000  | 0.483 (1.00) | 0.1752  | 0.1719  | 0.778 (1.00) |
| T6b within-page drift        | −0.0004 | −0.0004 | 0.720 (1.00) | −0.0019 | −0.0019 | 0.677 (1.00) |
| T7b deviant rate             | 23.5688 | 23.8003 | 0.382 (1.00) | 34.4531 | 33.9119 | 0.069 (1.00) |
| T7b deviant recurrence       | 0.0764  | 0.0756  | 0.839 (1.00) | 0.0582  | 0.0572  | 0.627 (1.00) |
| T7b slip share               | 0.2786  | 0.2811  | 0.670 (1.00) | 0.2816  | 0.2818  | 0.962 (1.00) |
| E1 line step                 | 0.0147  | 0.0124  | 0.651 (1.00) | 0.0153  | 0.0152  | 0.995 (1.00) |
| E1 paragraph step            | −0.0137 | −0.0068 | 0.376 (1.00) | 0.0000  | −0.0053 | 0.473 (1.00) |
| E1 within-line decay         | −0.0075 | −0.0068 | 0.828 (1.00) | 0.0004  | −0.0018 | 0.614 (1.00) |
| E1 within-paragraph decay    | 0.0010  | −0.0087 | 0.214 (1.00) | 0.0392  | 0.0364  | 0.723 (1.00) |

## S8. Definitions of the 40 phenotype statistics

**Profile statistics** (Section 3 of the main text):

- hapax share and type-token ratio;
- Zipf slope of the 200 most frequent types;
- mean word length;
- boundary MI: excess MI between the last glyph of a word and the first glyph of the next inside lines, against a within-line permutation null;
- word-pair MI and distance-2 word MI;
- line-break MI and line-break boundary MI, with the boundary MI at equal  $n$ ;
- space predictability: information gain about word boundaries from the flanking glyph pair;
- lexicon test: share of held-out real words and of glyph-trigram samples attested in the training vocabulary;
- conditional glyph entropies  $h_2$ – $h_4$ ;
- lzma bits per character;
- page MI between page identity and word.

**Mechanism battery:** T1 page-conditional line-break ratio; T2a final-form excess (margin-bound, paragraph-final); T2b slack ratio (stars section); T2c dittography ratio; T3a zero replication (250-token chunks); T3b slot consistency and monotone share; T3c positional entropy (final, initial); T4 any periodicity peak; T5 step/smooth prediction-error ratio; T6a near and exact recency indices; T6b slope of line similarity over line distance 1–8; T7b deviant rate, recurrence share and slip share.

**E1:** graded line step, paragraph step, graded within-line decay, within-paragraph decay.

**Tolerance:**  $|G - V| \leq 2\sqrt{(\text{SE}_V^2 + \text{SD}_G^2)} + 0.01|V|$ .  $\text{SE}_V$  is a leave-one-quiere-out jackknife over the test half;  $\text{SD}_G$  is taken over three generator seeds. T4 passes if at most one seed shows a peak.

## S9. Implementation notes

- **Page-conditional T1 estimator.** Per page, the excess plug-in MI over 200 within-page re-pairings, for the break pairs and for consecutive blocks of within-line pairs of the same size, averaged over pages with weights equal to the number of break pairs. The pooled estimators failed a known-truth simulation in which each page of Latin received a different letter substitution; this estimator returned identical values with and without substitution.
- **E1 expectations.** Position classes: offset from the line start (1 or  $\geq 2$ ), offset from the line end (1 or  $\geq 2$ ), paragraph-final line or not. The binary expectation is the exact mean over pairs of distinct eligible tokens of the page in the same class pair.
- **E5 null.** For each type, the expected same-page pairs are  $\sum_g C(n_{tg}, 2) \cdot \sum_{p \in g} N_p(N_p - 1) / (N_g(N_g - 1))$ , with groups  $g = \text{hand} \times \text{section}$ .
- **E7 coder.** 32-bit Witten–Neal–Cleary arithmetic coding over integer transition counts. The end-of-line suppression below 60% of the target length and the stop at 140% are deterministic and replicated by the receiver.
- **E8 coder.** The same coder drives the observable choices of the E3 generator. Floating-point probabilities are quantised to integer counts (resolution  $2^{-16}$ , minimum 1) identically in all conditions. The latent choices come from a key PRNG shared by sender and receiver: page tilt, line drift, route, exact or near reuse, final-word candidates, slips, and the intended form of slipped tokens. The receiver re-runs the generator with the key and re-encodes each observed choice. A token reached at the maximum length ends without a choice, which leaves the output distribution unchanged.